

Towards five-state minimal-time firing squads: a pumping lemma, a speed-1/3 barrier, and certified bounds

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Abstract

Whether the firing squad synchronization problem admits a minimal-time solution with five states has been open since Mazoyer’s six-state solution of 1987. We develop general tools for minimal-time solutions and measure how far they reach on this question. (i) All lines are encoded by one half-line diagram and one reflected triangle per length, and every triangle is a function of two anti-diagonals of the half-line. (ii) We prove two necessary conditions on the half-line alone, valid for any number of states: a pumping lemma, which forces the non-periodic part of the half-line to reach the line $i = t/3$ (a speed-1/3 barrier), and a left-border lemma, which bounds the periodic stretches of the anti-diagonals near the left border and hence forbids periodic behaviour in any wedge at the border. (iii) A propositional encoding of the bounded problem, validated against Mazoyer’s solution extracted from its Coq formalization, gives an LRAT-certified proof, checked in six seconds, that no four-state minimal-time solution exists, even without the left quiescence condition. (iv) For five states we exhibit explicit rules that synchronize every line of length at most 13 in minimal time and satisfy the half-line conditions up to length 40; the left-border lemma shows that their common half-line can never be completed to a solution. Combining both lemmas with certified solver calls, we show that every five-state minimal-time solution uses at least 15 distinct neighbourhoods on its half-line below the anti-diagonal $t + i = 78$. Random-probe estimates of the natural look-ahead search tree give about 27 nodes for four states but 10^{11} to 10^{19} for five, which quantifies why the question remains open.

Keywords: firing squad synchronization problem, cellular automata, state complexity, minimal time, SAT solving, proof certificates

1. Introduction

The firing squad synchronization problem (FSSP), posed by Myhill in 1957 and popularised by Moore [1], asks for a finite automaton that, placed in every cell of a line of arbitrary length n and started with all cells quiescent except a *general* at one end, drives all cells into a common *firing* state for the first time at the same instant. Goto [2] and Waksman [3] showed that this is possible in time $2n - 2$, which is optimal [1]. Since then the minimum number of states of a *minimal-time* solution has been a benchmark question in cellular automata theory [4, 5]: Waksman used 16 states, Balzer 8 [6], Gerken 7 [7] and Mazoyer 6 [8]. On the other side, Balzer [6] reported, and Sanders [9] confirmed with a corrected exhaustive search, that no 4-state minimal-time solution exists. Whether five states suffice has remained open since Mazoyer’s construction; it is regarded as the central open question on state complexity of the FSSP [5, 10]. Many six-state minimal-time solutions are known today [11, 12], and solutions with at most five states are known for restricted families of lengths, for instance $n = 2^k$, some of them in non-minimal time [13, 14, 15], but no five-state minimal-time solution has been found, nor has its existence been refuted.

This paper. We do not settle the five-state question. We develop tools that apply to minimal-time solutions with any number of states, use them to give a certified proof of the four-state lower bound that is produced and checked within a minute, and measure precisely how far the same tools reach for five states.

- (i) *Agreement and reflection* (Section 2). The line of length n coincides with the half-line below the anti-diagonal $t + i = 2n - 2$; hence all lines are described by a single half-line diagram C_∞ plus one *reflected triangle* R_n per length, and R_n is a function of two anti-diagonals of C_∞ only (Lemmas 2.2 and 3.1).
- (ii) *Two barriers* (Section 3). In every minimal-time solution the input words of the reflected triangles of any two lengths $4 \leq n < n'$ differ on an explicit cone of about $n/2$ cells (Lemma 3.3). This is a constraint on the half-line alone, valid for *all* pairs of lengths. It implies that the non-periodic part of the half-line always reaches the line $i = t/3$ (Theorem 3.4), an a priori explanation of the speed-1/3 signal present in the classical constructions. At the other end, the first chains of the reflected triangle form a finite-state system that must fire the cells near the left border exactly when it reaches them; hence the input anti-diagonals cannot be periodic over long stretches near the left border

(Lemma 3.5), and the half-line cannot be periodic in any wedge at the border (Theorem 3.6).

- (iii) *A validated encoding and a certified four-state bound* (Sections 4 and 5). We encode the bounded problem, together with the consequences above, into propositional logic, validate the encoding against Mazoyer’s solution extracted mechanically from Duprat’s Coq formalization [16], and obtain an LRAT certificate, checked in 6 seconds, that no four-state minimal-time solution exists (Theorem 5.1); the statement does not need the left quiescence condition $\delta(*, L, L) = L$.
- (iv) *The five-state frontier* (Section 6). We exhibit explicit five-state rules that synchronize every line of length at most 13 in minimal time, satisfy all pumping constraints up to length 40 and never fire on the half-line up to anti-diagonal 78; one of them ends with a uniform configuration for every length up to 12 (Theorem 6.1). So every refutation of five-state solutions must use lines of length at least 14. Their common half-line, however, is excluded by Lemma 3.5 at length 518 (Corollary 6.2). Combining the two lemmas with certified solver calls, we classify the half-lines of low complexity: every five-state solution uses at least 15 distinct neighbourhoods on its half-line below anti-diagonal 78 (Proposition 6.4). The difficulty lies in the half-line: once the transitions used on the half-line are fixed, the completion problem is usually decided in under a second, whereas random-probe estimates of the natural look-ahead search tree grow from about 27 nodes for four states to 10^{11} – 10^{19} for five.

Related work. Surveys of the FSSP are [4, 5]; a classical textbook exposition is [17]. Mazoyer [18] studied the structure of optimal solutions and the information flow they require; Settle and Simon [19] obtained small non-minimal-time solutions. Beyond Balzer and Sanders, lower bounds on the number of states are known for variants: no three-state solution and no four-state symmetric minimal-time solution exist for rings [20]; for further restricted classes see [10]. Computer searches have been used to find new solutions [11] and to generate families of solutions by local simulations [12]. SAT solvers have been applied to many extremal problems in combinatorics with certified unsatisfiability proofs [21, 22]; we are not aware of previous SAT-based work on the state complexity of the FSSP.

2. Preliminaries

2.1. The problem

Let S be a finite set of *states* containing three distinguished elements: the *quiescent* state L , the *general* G and the *firing* state F ; these are pairwise distinct. We write $W := S \setminus \{F\}$ for the set of *working* states and use a symbol $*$ $\notin S$ for the border. A *rule* is a map

$$\delta : (W \cup \{*\}) \times W \times (W \cup \{*\}) \longrightarrow S$$

satisfying the quiescence conditions

$$\delta(L, L, L) = \delta(L, L, *) = L. \quad (1)$$

(Transitions out of F and transitions whose neighbourhood contains F play no role below and are left unspecified; we do *not* require $\delta(*, L, L) = L$, which only strengthens the non-existence statements.)

For $n \geq 1$ the *line of length n* is the space-time diagram $C_n : \mathbb{N} \times \{1, \dots, n\} \rightarrow S$ defined by $C_n(0, 1) = G$, $C_n(0, i) = L$ for $2 \leq i \leq n$, and

$$C_n(t+1, i) = \delta(C_n(t, i-1), C_n(t, i), C_n(t, i+1)),$$

with the convention $C_n(t, 0) = C_n(t, n+1) = *$; the diagram is only considered up to the first time at which some cell is in state F . The rule *synchronizes the line of length n at time T* if $C_n(T, i) = F$ for all $1 \leq i \leq n$ and $C_n(t, i) \neq F$ for all $t < T$ and all i .

It is classical [1, 3, 6] that a rule which synchronizes *every* line cannot synchronize the line of length $n \geq 2$ before time $2n - 2$ (see Remark 2.4). A *minimal-time solution* is a rule that synchronizes the line of length n at time exactly $2n - 2$ for every $n \geq 2$. (For $n = 1$ one usually asks $\delta(*, G, *) = F$; this transition is used by no other line and is irrelevant here.) A *k -state solution* is one with $|S| = k$.

More generally, for a set $\mathcal{N} \subseteq \{2, 3, \dots\}$ of lengths, a rule is *$\mathcal{N}$ -minimal* if it synchronizes the line of length n at time $2n - 2$ for every $n \in \mathcal{N}$. A minimal-time solution is $\mathcal{N}$ -minimal for every $\mathcal{N}$. Since there are finitely many rules with k states, k -state minimal-time solutions do not exist if and only if there is a finite N such that no k -state rule is $\{2, \dots, N\}$ -minimal; a non-existence proof may therefore consist of a finite family of necessary conditions that no k -state rule satisfies. This is how we prove the four-state bound (Theorem 5.1); for five states we study how large such a family must be.

2.2. The half-line and the agreement lemma

Let $C_\infty : \mathbb{N} \times \{1, 2, \dots\} \rightarrow S$ be the diagram of the half-line: $C_\infty(0, 1) = G$, $C_\infty(0, i) = L$ for $i \geq 2$, and the same recurrence with a border on the left only.

Lemma 2.1 (light cone). *For every rule, every n and every (t, i) with $t < i - 1$ we have $C_n(t, i) = C_\infty(t, i) = L$.*

Proof. By induction on t . For $t = 0$ this is the initial condition. Let $t \geq 1$ and $t < i - 1$. The three cells $(t - 1, i - 1)$, $(t - 1, i)$, $(t - 1, i + 1)$ satisfy $t - 1 < (i - 1) - 1$, hence are in state L by induction, except that the right neighbour of cell n of C_n is the border. By (1) the new state is L in both cases. $\square$

Lemma 2.2 (agreement). *For every rule and every $n \geq 2$: $C_n(t, i) = C_\infty(t, i)$ whenever $1 \leq i \leq n$ and $t + i \leq 2n - 2$.*

Proof. Induction on t ; the case $t = 0$ is clear. Let $t \geq 1$ and $t + i \leq 2n - 2$. If $i = n$ then $t \leq n - 2 < i - 1$ and both sides equal L by Lemma 2.1. If $i < n$, the three neighbours $(t - 1, j)$, $j \in \{i - 1, i, i + 1\}$, satisfy $(t - 1) + j \leq t + i \leq 2n - 2$ and $j \leq n$, so they agree by induction (for $j = 0$ both are the border). Hence the new states agree. $\square$

Thus the right border of the line of length n is first “felt” at the cell $(n - 1, n)$, and its influence is confined to the *reflected triangle*

$$R_n := \{(t, i) : 1 \leq i \leq n, t + i \geq 2n - 1, t \leq 2n - 2\}.$$

Lemma 2.3 (return chain). *Let δ synchronize the lines of lengths n and $n + 1$ at times $2n - 2$ and $2n$ respectively. Then*

$$C_n(2n - 1 - i, i) \neq C_\infty(2n - 1 - i, i) \quad (1 \leq i \leq n).$$

*In particular the front $s_i := C_\infty(i - 1, i)$ satisfies $s_{n-1} \neq L$, and $\delta(s_{n-1}, L, *) \neq \delta(s_{n-1}, L, L)$.*

Proof. For $i = 1$: $C_n(2n - 2, 1) = F$ while, by Lemma 2.2 applied to the line of length $n + 1$ (note $(2n - 2) + 1 \leq 2(n + 1) - 2$), $C_\infty(2n - 2, 1) = C_{n+1}(2n - 2, 1) \neq F$ because $2n - 2 < 2n$. Suppose the claim holds for some $i < n$. The cell $(2n - 1 - i, i)$ is computed from the cells $(2n - 2 - i, j)$, $j = i - 1, i, i + 1$, whose coordinate sums are $2n - 3$, $2n - 2$ and $2n - 1$. By Lemma 2.2 the first two are the same in C_n and C_∞ (for $i = 1$ the left neighbour is the border in both diagrams). Since the results differ, the right neighbours differ:

$C_n(2n-2-i, i+1) \neq C_\infty(2n-2-i, i+1)$, which is the claim for $i+1$. For $i = n$ we get $C_n(n-1, n) = \delta(s_{n-1}, \mathbf{L}, *) \neq C_\infty(n-1, n) = \delta(s_{n-1}, \mathbf{L}, \mathbf{L}) = s_n$, which is impossible if $s_{n-1} = \mathbf{L}$ by (1). $\square$

Remark 2.4. Suppose a rule synchronizes every line, the line of length n at some time $T \leq 2n-3$. By Lemma 2.2, $C_\infty(T, 1) = C_n(T, 1) = \mathbf{F}$, hence $C_m(T, 1) = \mathbf{F}$ for every $m \geq n$. For $m > T+1$ the cell m is still quiescent at time T by Lemma 2.1, so the line of length m has a firing cell at time T without firing entirely: a contradiction.

Applying Lemma 2.3 to all n shows that in a minimal-time solution every cell i leaves $\mathbf{L}$ exactly at time $i-1$ (the front moves at speed one) and, since $C_\infty(t, i) = C_n(t, i)$ for n large,

$$C_\infty(t, i) \neq \mathbf{F} \quad \text{for all } t \geq 0, i \geq 1. \quad (2)$$

3. Reflected triangles and a pumping lemma

3.1. Relative coordinates

Fix $n \geq 2$. For a cell (t, i) of the line of length n put $\tau := t - (n-1)$ (time elapsed since the front reached the right end) and $\kappa := n - i$ (distance to the right end). Then

$$R_n = \{(\tau, \kappa) : 0 \leq \kappa \leq \tau \leq n-1\},$$

the right border sits at $\kappa = -1$ and the left border at $\kappa = n$. A cell $(\tau, \kappa) \in R_n$ is computed from $(\tau-1, \kappa+1), (\tau-1, \kappa), (\tau-1, \kappa-1)$; each of these is either in R_n , or a border, or satisfies $\tau' - \kappa' \in \{-1, -2\}$, i.e. lies on one of the anti-diagonals $t+i = 2n-2$, $t+i = 2n-3$, where C_n coincides with C_∞ by Lemma 2.2. We call

$$J_n(\kappa) := (C_\infty(n-3+\kappa, n-\kappa), C_\infty(n-2+\kappa, n-\kappa)), \quad 0 \leq \kappa \leq n-1,$$

the *input word* of R_n ; its two components are the *lower* ($t+i = 2n-3$) and *upper* ($t+i = 2n-2$) input cells.

Lemma 3.1 (determinacy). *There is a single partial map Ψ , depending on δ only, such that for every $n \geq 2$ and every $(\tau, \kappa) \in R_n$, $C_n(n-1+\tau, n-\kappa)$ is obtained by evaluating Ψ at (τ, κ) on the input word J_n with borders at $\kappa = -1$ and $\kappa = n$. In particular the value only depends on the input cells and borders in the backward dependency cone of (τ, κ) .*

Proof. The recurrence in relative coordinates does not mention n ; only the position of the left border does. $\square$

For $(\tau, \kappa) \in R_n$ let $\text{cone}_n(\tau, \kappa)$ be the set of input cells, i.e. of pairs consisting of a position κ' and one of the two anti-diagonals, that are reached from (τ, κ) by repeatedly passing to the three predecessors, stopping at input cells and borders; say that the cone is *right-anchored* if it does not reach the left border $\kappa' = n$.

Lemma 3.2. *For $n \geq 4$ the cone of the right end at the firing time, $\text{cone}_n(n-1, 0)$, is right-anchored and consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \leq \kappa' \leq \lfloor n/2 \rfloor$, and, when n is odd, the lower input cell at $\kappa' = (n+1)/2$.*

Proof. A predecessor step changes τ by -1 and κ by at most one, and input cells satisfy $\tau' = \kappa' - 1$ or $\kappa' - 2$. Starting from $(n-1, 0)$, a path that reaches an input cell at position κ' after s steps satisfies $\kappa' \leq s$ and $n-1-s = \kappa' - 1$ or $\kappa' - 2$; hence $2\kappa' \leq n+1$ (upper cells: $2\kappa' \leq n$). Conversely, the upper cell at a position κ' with $2\kappa' \leq n$ is reached by moving left κ' times and then staying put, and the lower cell at a position $\kappa' \geq 1$ with $2\kappa' \leq n+1$ by moving left $\kappa' - 1$ times, staying put down to the cell $(\kappa' - 1, \kappa' - 1)$ and moving left once more; all intermediate cells satisfy $\tau \geq \kappa$ and lie in R_n . The lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n . The largest position reached is $\lceil n/2 \rceil \leq n-1$, and a path can reach $\kappa = n$ only after n left moves, i.e. at a negative time, so the left border is never reached; the right border only contributes the constant symbol $*$. (The same set is also computed mechanically in the accompanying code, file `pumping.py`.) $\square$

Lemma 3.3 (pumping). *Let δ be a minimal-time solution and $4 \leq n < n'$. Then J_n and $J_{n'}$ differ on $\text{cone}_n(n-1, 0)$.*

Proof. Suppose they agree there. By Lemmas 3.1 and 3.2 the relative cell $(n-1, 0)$ of $R_{n'}$ (which belongs to $R_{n'}$ since $n-1 \leq n'-1$, and whose cone is the same set of input positions, far from the left border $\kappa = n' > n$) has the same value as the relative cell $(n-1, 0)$ of R_n , namely $C_n(2n-2, n) = \mathbf{F}$. Thus $C_{n'}(n' + n - 2, n') = \mathbf{F}$ although $n' + n - 2 < 2n' - 2$, contradicting minimality for n' . $\square$

The point of Lemma 3.3 is that it constrains the *half-line diagram only*: no reflected triangle has to be computed. It is the source of the constraint family $\text{PUMP}(n, n')$ used in Section 4. As a sanity check, we verified mechanically that Mazoyer's six-state solution satisfies $\text{PUMP}(n, n')$ for all $4 \leq n < n' \leq 120$.

3.2. Consequence: a speed-1/3 barrier

Write the half-line diagram in the frame of the front: $\Phi_t(j) := C_\infty(t, t + 1 - j)$ for $0 \leq j \leq t$ (the cell at *depth* j behind the front at time t). By the recurrence,

$$\Phi_{t+1}(j) = \delta(\Phi_t(j), \Phi_t(j-1), \Phi_t(j-2)) \quad (0 \leq j \leq t), \quad (3)$$

with $\Phi_t(-1) = \Phi_t(-2) = \mathbf{L}$: in this frame the half-line is a *one-sided* automaton in which information only travels to larger depths, at most two depths per step. In particular every depth-row $t \mapsto \Phi_t(j)$ is eventually periodic. In these coordinates $J_n(\kappa) = (\Phi_{n-3+\kappa}(2\kappa-2), \Phi_{n-2+\kappa}(2\kappa-1))$.

Theorem 3.4 (speed-1/3 barrier). *Let δ be a minimal-time solution. Every depth-row $t \mapsto \Phi_t(j)$ is eventually periodic; let $T(j)$ be the time from which it is periodic. Then for every $n \geq 4$ there is a depth $j \leq n$ with $T(j) > n + \frac{j}{2} - \frac{5}{2}$. Consequently $\limsup_{j \rightarrow \infty} T(j)/j \geq 3/2$: along infinitely many depth-rows the transient part extends at least to the line $i = t/3$ of the original coordinates.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), \mathbf{L}, \mathbf{L})$ and is eventually periodic; by (3), row j is a finite-state system driven by the eventually periodic rows $j-1$, $j-2$, hence eventually periodic, with some period P_j . Fix $n \geq 4$ and let P be a common multiple of $P_0, \dots, P_{n+1}$. If every input cell of $\text{cone}_n(n-1, 0)$ were read in the periodic regime of its row, i.e. $n-3+\kappa \geq T(2\kappa-2)$ and $n-2+\kappa \geq T(2\kappa-1)$ for all cone positions κ , then $J_{n+P} = J_n$ on the cone, contradicting Lemma 3.3. Cone positions satisfy $\kappa \leq \lceil n/2 \rceil$, so the violated inequality concerns a depth $j \in \{2\kappa-2, 2\kappa-1\}$ with $j \leq n$ and $T(j) > n-3+\kappa \geq n + \frac{j}{2} - \frac{5}{2}$. If $T(j) \leq \alpha j + \beta$ for all j with $\alpha < 3/2$, then for n large and every $j \leq n$ we would have $(\alpha - \frac{1}{2})j \leq (\alpha - \frac{1}{2})n < n - \beta - \frac{5}{2}$, i.e. $T(j) \leq n + \frac{j}{2} - \frac{5}{2}$, a contradiction. Finally a cell of depth j at time $T(j)$ sits at position $T(j) + 1 - j$, which is at least about $T(j)/3$ when $T(j) \geq 3j/2$. $\square$

Theorem 3.4 explains a posteriori why every minimal-time construction contains structure reaching the line $i = t/3$, whose crossing with the anti-diagonal $t+i = 2n-2$ is exactly the far end of $\text{cone}_n(n-1, 0)$: the speed-1/3 signal emitted by the general in the constructions of Waksman and Balzer [3, 6] lies exactly on it, and in Mazoyer’s solution the periodic zone behind the front is bounded by a line of speed 1/2; indeed we checked that for $1 \leq j < 700$ the depth-row j is 3-periodic from time $T(j) = 2j+2$ when j is even or $j=1$, and from $T(j) = 2j+1$ when $j \geq 3$ is odd. The theorem also explains the failure of many of the “lucky” partial rules found by the solvers

(Section 6): their half-line diagrams are periodic behind the front far beyond the line $i = t/3$.

3.3. A second barrier, at the left border

Lemma 3.3 looks at the right end of the line. The left end gives a constraint of a different nature, again on the half-line only. Fix $n \geq 2$ and number the anti-diagonals of the line of length n from the input anti-diagonals on: for $c \geq -1$ and $1 \leq i \leq n$ with $c \leq i$ put

$$v_c(i) := C_n(2n - 2 + c - i, i),$$

the cell of the chain $t + i = 2n - 2 + c$ at position i . Chains -1 and 0 are the input anti-diagonals, on which $v_c(i) = C_\infty(2n - 2 + c - i, i)$ by Lemma 2.2; chains $c \geq 1$ lie in R_n , and chain 1 is the return chain of Lemma 2.3. The cell $(2n - 2 + c - i, i)$ has its three predecessors on the chains $c - 2$, $c - 1$ and c , so

$$v_c(i) = \delta(v_{c-2}(i-1), v_{c-1}(i), v_c(i+1)) \quad (c \geq 1), \quad (4)$$

where $v_{c-2}(0)$ and $v_c(n+1)$ stand for the border. The line of length n is synchronized at time $2n - 2$ if and only if $v_i(i) = \mathbf{F}$ for $1 \leq i \leq n$ and $v_c(i) \neq \mathbf{F}$ for $c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$, (4) becomes

$$u_c(y) = \delta(u_{c-2}(y), u_{c-1}(y + \varepsilon_c), u_c(y+1)), \quad \varepsilon_c = \begin{cases} 1, & c \text{ even,} \\ 0, & c \text{ odd,} \end{cases} \quad (5)$$

with $u_0(y) = C_\infty(2n - 2 - y, y)$ and $u_{-1}(y) = C_\infty(2n - 2 - y, y - 1)$, and $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n - 2 + \lceil c/2 \rceil - y$: it must be $\mathbf{F}$ if $\lceil c/2 \rceil = y$ and must differ from $\mathbf{F}$ if $\lceil c/2 \rceil < y$. Thus each *slice* $(u_1(y), u_2(y), \dots)$ is obtained from the slice $y+1$ by a sequential finite-state transducer that reads the components in increasing order and whose initial state is determined by the two input cells $u_{-1}(y), u_0(y)$: read from right to left, the reflected triangle is a cascade of finite-state transducers, the left-hand counterpart of the co-moving description (3) of the half-line.

Lemma 3.5 (left border). *Let δ have k states and synchronize the line of length n at time $2n - 2$. Let $s \geq 1$, $L \geq 1$ and $s + L \leq Y \leq n - s$, and suppose that the words*

$$\begin{aligned} y &\mapsto C_\infty(2n - 2 - y, y) & (s \leq y \leq Y) & \quad \text{and} \\ y &\mapsto C_\infty(2n - 3 - y, y) & (s \leq y \leq Y - 1) \end{aligned}$$

are L -periodic. Then $Y - s \leq (k - 1)^{2s} L$.

Proof. For $s \leq y \leq Y$ let $P_y := (u_1(y), \dots, u_{2s}(y))$. These are cells $y + \lfloor c/2 \rfloor \leq Y + s \leq n$ with $\lceil c/2 \rceil \leq s \leq y$, so they are defined, and for $y > s$ none of them is F. By (5), $u_c(y)$ with $1 \leq c \leq 2s$ is computed from components of index $< c$ of the same slice, from components of index $\leq c$ of slice $y + 1$, and from $u_{-1}(y), u_0(y)$; hence there is a map Θ , depending on δ only, with

$$P_y = \Theta(P_{y+1}, u_{-1}(y), u_0(y)) \quad (s+1 \leq y \leq Y-1).$$

Unfolding (5) for even c , $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y)$ depends only on $u_0(y)$ and P_{y+1} : there is a map E with $u_{2s}(y) = E(P_{y+1}, u_0(y))$ whenever $s \leq y \leq Y-1$. By hypothesis, u_0 is L -periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose $Y - s > (k-1)^{2s}L$. The pairs $(P_y, y \bmod L)$, $s+1 \leq y \leq Y$, take at most $(k-1)^{2s}L$ values, so there are $s+1 \leq b < a \leq Y$ with $P_a = P_b$ and $a \equiv b \pmod{L}$; put $d := a - b$. By downward induction, $P_y = P_{y+d}$ for $s+1 \leq y \leq b$: this holds for $y = b$, and if it holds for $y+1 \leq b$ then $P_y = \Theta(P_{y+1}, u_{-1}(y), u_0(y)) = \Theta(P_{y+1+d}, u_{-1}(y+d), u_0(y+d)) = P_{y+d}$, because $y+d < a \leq Y$ and $L \mid d$. In particular $P_{s+1} = P_{s+1+d}$, and since $s+d \leq Y-1$ and $u_0(s+d) = u_0(s)$,

$$u_{2s}(s+d) = E(P_{s+1+d}, u_0(s+d)) = E(P_{s+1}, u_0(s)) = u_{2s}(s) = \text{F}.$$

But $\lceil 2s/2 \rceil = s < s+d$, so $u_{2s}(s+d)$, the cell $2s+d$ at time $2n-2-d$, is not F: a contradiction. $\square$

In words: a band of $2s$ chains that crosses a long periodic stretch of the input anti-diagonals is a finite-state system, so it cannot time the firing of the cell $2s$, which it must trigger exactly when it arrives there. The border itself is harmless (cell 1 fires because the return chain meets the border), which is why the lemma starts at $s \geq 1$ and pays a factor exponential in s .

Theorem 3.6 (left-border barrier). *Let δ be a minimal-time solution. There are no $s \geq 1$, $\alpha > 0$, t_0 and periods $p, q \geq 1$ such that $C_\infty(t+p, i) = C_\infty(t, i)$ and $C_\infty(t, i+q) = C_\infty(t, i)$ whenever the cells involved lie in the wedge $\Omega := \{(t, i) : s \leq i \leq \alpha t, t \geq t_0\}$.*

Proof. Suppose the contrary and put $L := pq$. If (t, i) and $(t-L, i+L)$ lie in Ω , then so do the intermediate cells $(t-L, i+jq)$ and $(t-L+jp, i)$, and $C_\infty(t-L, i+L) = C_\infty(t-L, i) = C_\infty(t, i)$. For the anti-diagonals $t+i = 2n-2$ and $t+i = 2n-3$ this makes the words of Lemma 3.5 L -periodic on $[s, Y_n]$ for some Y_n with $Y_n \geq \frac{\alpha}{1+\alpha} 2n - C$, where C depends

on α, L, t_0 only. For n large, $\min(Y_n, n - s) - s > (k - 1)^{2s}L$, contradicting Lemma 3.5. $\square$

Theorems 3.4 and 3.6 complement each other: the half-line of a minimal-time solution has non-periodic structure reaching the line $i = t/3$ behind the front, and non-periodic structure in every wedge at the left border. In the constructions of Waksman and Balzer the latter is the family of signals of speeds $1/(2^j - 1)$, which accumulate at the border; in Mazoyer's solution it is the family of recursive division points. We checked Lemma 3.5 mechanically on Mazoyer's rule for $n \leq 400$, $s \leq 3$, $L \leq 6$: its periodic stretches near the left border have length at most 6, far below the bound. Unlike Lemma 3.3, Lemma 3.5 needs long periodic stretches (for five states, more than $16L$ cells when $s = 1$ and more than $256L$ cells when $s = 2$), hence long lines, but it applies to any half-line that can be computed far enough, which we exploit in Section 6.

4. Propositional encoding

Fix k (the number of states), write $W = \{\mathbf{L}, \mathbf{G}, a_1, \dots, a_{k-3}\}$ and let $\mathcal{N}$ be a finite set of lengths. We build a CNF formula $\text{MT}_k(\mathcal{N}; M, N')$ that is satisfiable whenever a k -state minimal-time solution exists. All constraint families below are either the definition of $\mathcal{N}$ -minimality or consequences of Lemmas 2.2, 2.3, 3.1 and 3.3 and (2); none relies on unproved assumptions about the shape of solutions.

Variables. For every neighbourhood $e = (x, y, z) \in (W \cup \{*\}) \times W \times (W \cup \{*\})$ other than $(*, y, *)$, one-hot variables $T_{e,d}$, $d \in S$, encode $\delta(e) = d$; the two quiescence transitions (1) are fixed by unit clauses. For every cell whose state is not a constant, one-hot variables $X_{c,s}$, $s \in W$, encode its state (all cells whose state is used before the firing time are in W).

Transition clauses. For a cell c with neighbours c_1, c_2, c_3 (cells, constants $\mathbf{L}, \mathbf{G}$, or the border) and for all (x, y, z) in the respective domains and $d \in W$ we add

$$\begin{aligned} &\neg X_{c_1,x} \vee \neg X_{c_2,y} \vee \neg X_{c_3,z} \vee \neg T_{(x,y,z),d} \vee X_{c,d}, \\ &\neg X_{c_1,x} \vee \neg X_{c_2,y} \vee \neg X_{c_3,z} \vee \neg T_{(x,y,z),\mathbf{F}} \end{aligned}$$

(the second clause forbids firing). For a cell that must fire we add instead $\neg X_{c_1,x} \vee \neg X_{c_2,y} \vee \neg X_{c_3,z} \vee T_{(x,y,z),\mathbf{F}}$. Literals of constant neighbours are omitted.

Half-line. The cells (t, i) of C_∞ with $1 \leq i \leq t + 1$ and $t + i \leq M$ receive variables and transition clauses; cells with $i > t + 1$ are the constant $\mathbf{L}$ (Lemma 2.1). Every such cell is forbidden to fire, by (2).

Lines. For every $n \in \mathcal{N}$ (with $2n - 2 \leq M$) the cells of R_n with $t \leq 2n - 3$ receive variables; their neighbours outside R_n are the corresponding half-line cells (Lemma 2.2). All of them are forbidden to fire and the n cells at time $2n - 2$ are required to fire.

Return chains. For $n \in \mathcal{N}$ with $2n - 1 \leq M$ we add the implied constraints $C_n(2n - 1 - i, i) \neq C_\infty(2n - 1 - i, i)$ of Lemma 2.3 (their derivation only uses (2) at the cell $(2n - 2, 1)$, which is part of the encoding).

Pumping. For all $4 \leq n < n' \leq N'$ (with $2N' - 2 \leq M$) we add one clause stating that J_n and $J_{n'}$ differ on $\text{cone}_n(n - 1, 0)$, introducing an auxiliary “difference” variable for every pair of non-constant input cells (Lemma 3.3).

Determinacy links. For $n < n'$ in $\mathcal{N}$ and $p \geq 0$ an auxiliary variable $E_{n,n'}(p)$ is made equivalent to “ J_n and $J_{n'}$ coincide on the positions $0, \dots, p$ ”. For every relative cell (τ, κ) of R_n whose dependency cone is right-anchored and contained in the positions $0, \dots, p$ we add $E_{n,n'}(p) \Rightarrow (R_n(\tau, \kappa) = R_{n'}(\tau, \kappa))$, and, for the firing cells $\tau = n - 1$, the clause $\neg E_{n,n'}(p)$ (Lemmas 3.1 and 3.3). These clauses are implied by the others but allow the solver to transfer information between lines.

Symmetry breaking. Permuting the auxiliary states maps solutions to solutions. We require that, in the time-major enumeration of the encoded half-line cells, the first occurrence of a_j precedes the first occurrence of a_{j+1} ; every solution can be renamed to satisfy this. For $k = 5$ this halves the search space.

Soundness checks. Besides the proofs above, every constraint family was tested against an independent ground truth: Mazoyer’s six-state solution, extracted mechanically from Duprat’s Coq formalization [16], satisfies the formula $\text{MT}_6(\mathcal{N}; 78, 40)$ for $\mathcal{N} = \{3, \dots, 12\}$ with return chains, pumping and links (the formalization does not cover $n = 2$), and it satisfies $\text{MT}_6(\{3, \dots, 14\}; 26)$ with symmetry breaking under exactly one of the 3! renamings of its auxiliary states, as it must. Conversely, every model found by the solvers was decoded and re-simulated by an independent brute-force checker (`fsspcheck.c`), which confirmed minimal-time synchronization for all encoded lengths.

largest length N	5	6	7	8	9
partial rules synchronizing $2, \dots, N$	828,670	64,299	536	27	0

Table 1: Four states: number of partial rules (restricted to the transitions they use) that synchronize every line of length $2, \dots, N$ in minimal time, by exhaustive enumeration ($1.09 \cdot 10^8$ search nodes).

5. Four states: a certified proof

All formulas are produced by the generator `gencnf.py`; models are decoded into rule tables and re-simulated by the independent brute-force checker `fsspcheck.c`. Solvers: Kissat 4.0.4 [23] and CaDiCaL 3.0.1 [24] (or CaDiCaL 1.9.5 through PySAT [25] for incremental runs). Times are single-core wall-clock times on a cloud virtual CPU.

Theorem 5.1. *No rule with four states $\{L, G, a, F\}$ satisfying (1) synchronizes all lines of lengths $2, \dots, 9$ in minimal time. In particular there is no four-state minimal-time solution, even without the condition $\delta(*, L, L) = L$.*

Proof. By Lemmas 2.2 and 2.3 and (2) every such rule yields a model of $MT_4(\{2, \dots, 9\}; 16)$ (for $k = 4$ there is a single auxiliary state and no symmetry breaking). CaDiCaL proves this formula unsatisfiable in 51 s and emits an LRAT proof of 177 MB (41 MB compressed), which `lrat-check` of the `drat-trim` suite [26] verifies in 5.7 s. The SHA-256 digests of the formula and of the proof are listed in Appendix B. $\square$

The bound 9 is sharp for the plain formula: $MT_4(\{2, \dots, 8\}; 14)$ is satisfiable, and an independent depth-first enumeration (program `fsspdfs.c`, which fills the table lazily in the order in which transitions are needed) finds the numbers of partial rules listed in Table 1, counted on the transitions they use; it reaches zero exactly at the lengths $2, \dots, 9$, in agreement with the solver. With the half-line extended to anti-diagonal 78 the lengths $2, \dots, 8$ already suffice. The pumping constraints alone are weak for four states: together with the half-line up to anti-diagonal 78, the lengths $2, \dots, 7$ remain satisfiable.

6. Five states: the frontier

6.1. Lucky rules

For five states the table has 94 free transitions with five values each, against 45 with four values for four states: the search space grows by a

instance ($k = 5$)	result	time
$\text{MT}_5(\{2, \dots, 10\}; 18)$	SAT, fails at $n = 11$	10.8 s
$\text{MT}_5(\{2, \dots, 12\}; 22)$	SAT, fails at $n = 13$	≈ 20 min
$\text{MT}_5(\{2, \dots, 9, 16\}; 30)$	SAT, fails at $n = 10$	23.6 s
$\text{MT}_5(\{2, \dots, 9, 20\}; 38)$	SAT, fails at $n = 10$	82.4 s
$\text{MT}_5(\{2, \dots, 8\}; 78, 40)$	SAT, fails at $n = 9$	68.7 s
$\text{MT}_5(\{2, \dots, 10\}; 78, 40)$	SAT, fails at $n = 11$	755.6 s
$\text{MT}_5(\{2, \dots, 12\}; 78, 40) + \text{links} + \text{uniform pre-firing } A^n$	SAT, fails at $n = 13$	1930 s

Table 2: Five-state instances. $\text{MT}_k(\mathcal{N}; M, N')$: lengths $\mathcal{N}$, half-line up to anti-diagonal M , pumping pairs up to N' .

factor of about 10^{38} . Table 2 lists representative instances. All of them are satisfiable, and in each case the decoded rule fails for some length that was not encoded; we call such rules *lucky*. Two kinds occur. Rules of the first kind have a *periodic* half-line behind the front (Figure 1(b)); they can even synchronize an isolated long line such as $n = 20$, but they violate the pumping lemma massively (155 of the pairs $4 \leq n < n' \leq 40$ for the rule shown). Rules of the second kind have a chaotic half-line which, without the half-line constraints, typically fires at some point. Once the half-line is extended and the pumping constraints are imposed, the solver is forced to produce half-lines with a genuine slow boundary: in the models we examined in detail—Figure 1(c) and the rules of Theorem 6.1, whose common half-line consists of a front of G's, a solid wake of G's and a region $A L A L \dots$ behind a boundary of speed exactly $1/3$ —a region delimited by a line of slope $1/3$ appears, as predicted by Theorem 3.4.

6.2. Five-state rules for all lengths up to thirteen

Theorem 6.1. *The five-state rules δ_{12} and δ_{13} of Tables A.4 and A.5 have the following properties.*

- (a) δ_{12} synchronizes the line of length n at time $2n-2$ for every $2 \leq n \leq 12$, and one step before firing all cells are in state A. It does not synchronize the line of length 13: a cell fires at time 23.
- (b) δ_{13} synchronizes the line of length n at time $2n-2$ for every $2 \leq n \leq 13$, and one step before firing all cells are in state A whenever $4 \leq n \leq 13$. It does not synchronize the line of length 14: a cell fires at time 21.

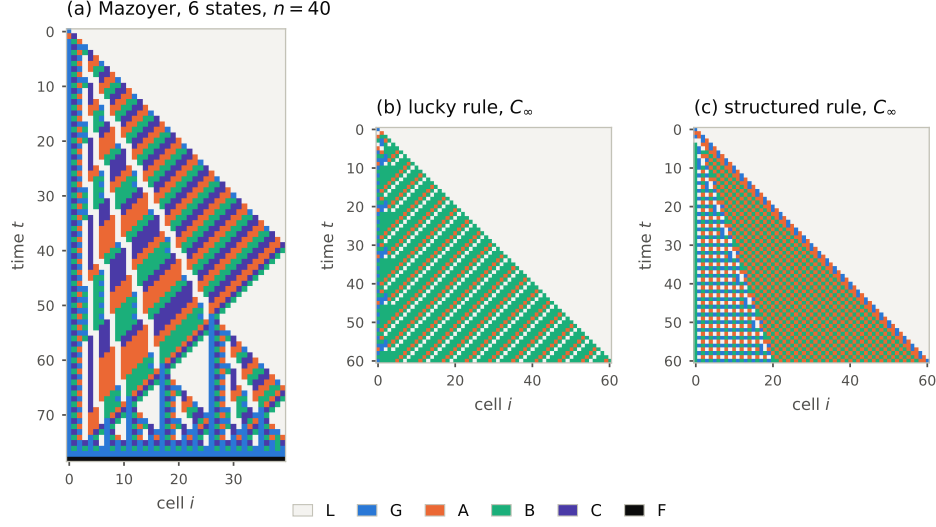


Figure 1: Space-time diagrams, time downwards. (a) Mazoyer's six-state solution on $n = 40$ cells; the last row is the firing row. (b) Half-line of a five-state rule that synchronizes the lengths $2, \dots, 9$ and 20 but violates the pumping lemma: its background is periodic. (c) Half-line of a five-state rule found under the pumping constraints: the boundary between the checkered region and the diagonal background has slope $1/3$.

(c) *The half-line diagrams of δ_{12} and δ_{13} coincide on all cells (t, i) with $t + i \leq 78$; these cells use 22 neighbourhoods besides (L, L, L) , on which the two rules agree, and none of them is a firing cell.*

(d) *Both rules satisfy $\text{PUMP}(n, n')$ for all $4 \leq n < n' \leq 40$.*

Proof. Each property is a finite computation on the explicit tables, carried out by direct simulation: (a), (b) and the failures at $n = 13$ and $n = 14$ with the checker `fsspcheck.c`, (c) with `check_halfline.py` and `germcompare.py`, and (d) with `pumping.py`, which recomputes the cones of Lemma 3.2 by breadth-first search (Appendix B). $\square$

The rule δ_{12} was found by Kissat on $\text{MT}_5(\{2, \dots, 12\}; 78, 40)$ with links and the uniform pre-firing constraint (Table 2); δ_{13} was found by the same solver after fixing the 22 half-line neighbourhoods of δ_{12} and adding the length 13, in 108s. The two rules differ in 57 of the remaining entries. Consequently, the necessary conditions used in this paper, restricted to lengths at most 13 (even together with the pumping constraints up to 40 and the half-line up to anti-diagonal 78), are satisfiable: a refutation of five-state

minimal-time solutions along these lines must involve lines of length at least 14, and the requirement “uniform configuration before firing”, satisfied by Mazoyer’s solution, does not help below length 12. The half-line shared by the two rules consists of a front of G’s, a solid wake of G’s and, behind a boundary of speed exactly $1/3$, a region $ALAL\cdots$ that is periodic in time and space. By the left-border barrier this architecture cannot be completed to a solution:

Corollary 6.2. *No rule that agrees with δ_{12} on the 22 neighbourhoods of Theorem 6.1(c) synchronizes the line of length 518 in minimal time. In particular no minimal-time solution has the half-line of δ_{12} and δ_{13} below anti-diagonal 78.*

Proof. Simulating the half-line with these 22 neighbourhoods (and (L, L, L)) only, no other neighbourhood occurs on the cells with $t + i \leq 1034$; hence every such rule has the same diagram C_∞ there. On the anti-diagonal $t + i = 1034$ the cells $2, \dots, 259$ are all in state A, and on $t + i = 1033$ the cells $2, \dots, 258$ are all in state L. Lemma 3.5 with $k = 5$, $n = 518$, $s = 2$, $L = 1$ and $Y = 259$ would give $257 \leq 4^4 = 256$. $\square$

So no choice of the remaining transitions can rescue this architecture. The lemma only bites at length 518, whereas the two rules themselves already fail at lengths 13 and 14.

The frontier is not tied to this architecture. The classification of Section 6.3 below left a few germs of complexity 16 undecided, and one of them leads to a second rule for all lengths up to thirteen, with a half-line of an entirely different kind (Figure 2).

Proposition 6.3. *The five-state rule δ'_{13} of Table A.6 synchronizes the line of length n at time $2n - 2$ for every $2 \leq n \leq 13$ and fails for $n = 14$ (a cell fires at time 22). Its half-line diagram uses 16 neighbourhoods besides (L, L, L) on all cells with $t + i \leq 10^4$, has no firing cell there, and satisfies $\text{PUMP}(n, n')$ for all $4 \leq n < n' \leq 800$; for $n \leq 5000$, $s \leq 3$ and $L \leq 24$ its input anti-diagonals have no stretch to which Lemma 3.5 applies.*

Proof. Direct simulation with the programs `fsspcheck.c`, `germcompare.py`, `germext.c` and `pumping.py`. $\square$

The rule was found by Kissat in 442s after fixing the 16 neighbourhoods and adding the lengths $2, \dots, 13$. Its half-line (Figure 2(b)) is chaotic: it has non-periodic structure everywhere, so it passes both barriers of Section 3 trivially, and the only arguments that apply to it are completion tests for longer lines, which are expensive.

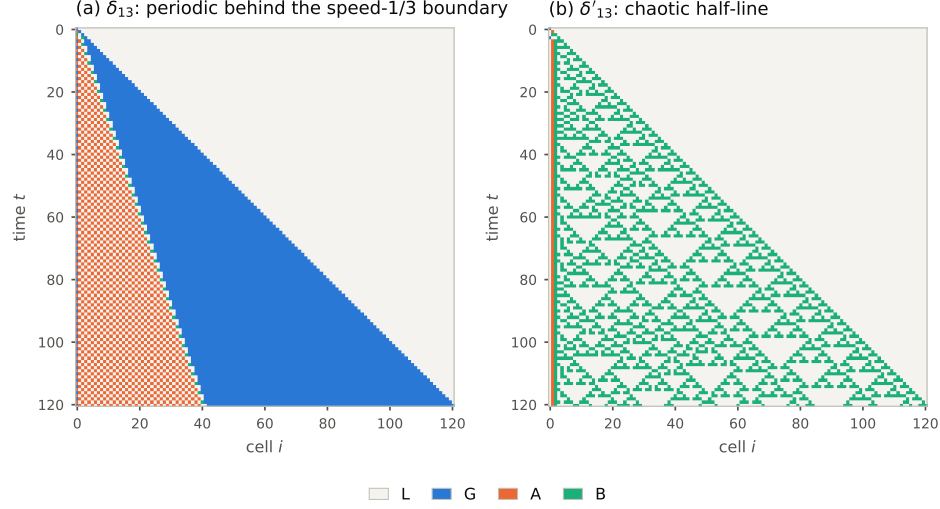


Figure 2: Half-lines of the two architectures that synchronize every line of length at most 13 (time downwards, $0 \leq t \leq 120$). (a) The common half-line of δ_{12} and δ_{13} : a front and a solid wake of G's and a checkered region of A's and L's behind a boundary of speed $1/3$; the checkered region is excluded by Lemma 3.5 (Corollary 6.2). (b) The half-line of δ'_{13} , determined by 16 neighbourhoods; its irregular pattern escapes both barriers.

6.3. Half-lines of low complexity

For a rule δ and a bound M , the *germ* of δ below M is the restriction of δ to the neighbourhoods $(C_\infty(t-1, i-1), C_\infty(t-1, i), C_\infty(t-1, i+1))$ of the cells (t, i) with $t \geq 1$ and $t+i \leq M$ of its half-line diagram, and its *complexity* $c_M(\delta)$ is the number of these neighbourhoods other than (L, L, L) . The germ determines C_∞ on all cells with $t+i \leq M$. The rules of Theorem 6.1 have $c_{78} = 22$ and Mazoyer's rule has $c_{78} = 57$.

Proposition 6.4. *Every five-state minimal-time solution δ satisfies $c_{78}(\delta) \geq 15$.*

Proof (computer-assisted). Let δ be a five-state minimal-time solution with $c_{78}(\delta) \leq 14$. Order the cells of C_∞ by anti-diagonal and, within an anti-diagonal, by increasing time; after exchanging the names A and B if necessary, A occurs before B in this order (if at all). By (2), Lemmas 2.3 and 3.3, the germ of δ below 78 produces no firing cell, never produces L at the front, and satisfies PUMP(n, n') for $4 \leq n < n' \leq 40$. The program `germdfs.c` enumerates all germs with these properties, with at most 14 neighbourhoods besides (L, L, L) and with A before B, by a depth-first search

complexity bound $c_{78} \leq$	14	16
search nodes of <code>germdfs.c</code>	$1.0 \cdot 10^8$	$1.5 \cdot 10^9$
germs (non-firing, front $\neq L$, PUMP to 40)	244	9,787
refuted by the completion test for $2, \dots, 10$ (LRAT)	230	8,824
then refuted by Lemma 3.5 or Lemma 3.3	14	929
remaining	0	34
refuted by Lemma 3.5 or Lemma 3.3 alone	218	$\geq 8,263$

Table 3: Classification of the half-line germs of low complexity (Section 6.3). The last row applies the half-line arguments to all germs, without completion tests (half-lines simulated up to anti-diagonal 3300 for $c_{78} \leq 14$ and 1300 for $c_{78} \leq 16$).

over the cells in the above order in which the value of a neighbourhood is chosen when it first occurs: $1.0 \cdot 10^8$ nodes, 244 germs. For 230 of them the formula $MT_5(\{2, \dots, 10\}; 78, 39)$ (without symmetry breaking, which the enumeration has already performed) together with the germ is unsatisfiable; a single formula in which a selector variable chooses the germ is refuted by CaDiCaL in 4s, with an LRAT proof (11 MB) checked by `lrat-check` in under a second. Each of the remaining 14 germs determines C_∞ far beyond anti-diagonal 78 (no new neighbourhood occurs up to anti-diagonal 3300), and on these extended diagrams Lemma 3.5 applies with $s = 1$: for each germ there are $n \leq 163$ and $L \leq 5$ such that the two input anti-diagonals of the line of length n are L -periodic on $[1, Y]$ with $Y - 1 > 16L$ (for instance $n = 28$, $L = 1$, $Y = 19$). Hence no rule containing one of the 244 germs is a minimal-time solution. $\square$

The same pipeline, run on the 9,787 germs of complexity at most 16 ($1.5 \cdot 10^9$ nodes), shows how the two kinds of arguments share the work (Table 3). The completion test for the lengths $2, \dots, 10$ refutes 8,824 germs (one LRAT proof of 2.0 GB, found in 556s and checked in 51s). Of the 963 survivors, 929 are refuted by Lemma 3.5 or Lemma 3.3 applied to their half-lines, each of which is determined by the germ far enough for the lemma to apply (we simulate up to anti-diagonal 10^4). Used alone, the half-line arguments refute 8,263 of the 9,787 germs, without any SAT call. Among the 34 germs that pass both filters are germs whose half-lines contain self-similar, Sierpiński-like regions, on which Lemma 3.5 has no grip, and germs whose periodic regions start too far from the border for the bound of the lemma to be reached within our simulation range.

6.4. Where the difficulty lies

The half-line determines everything. Take any of the rules found above, fix the transitions it uses on the half-line up to anti-diagonal 78 (between 39 and 50 transitions, counting (L, L, L)) and ask whether the remaining transitions can be completed so that all lengths $2, \dots, N$ synchronize. For the five-state germs of Table 2, Kissat decides this in less than one second: for the rule of Table 2 with lengths $2, \dots, 10$ the completion exists for $N = 8$ and not for $N = 12$; the other germs fail already for $N = 8$. The chaotic germ of Proposition 6.3 is an exception: its completions for $N = 12$ and $N = 13$ take 2 s and 442 s, and $N = 14$ is undecided after 600 s. As a sanity check on a genuine solution, fixing the 58 transitions that Mazoyer’s rule uses on its half-line up to anti-diagonal 78 (that is, $c_{78} = 57$ and (L, L, L)) leaves a completion problem for the lengths $3, \dots, 12$ that Kissat solves in 4 s. Conversely, the unsatisfiable core returned for such a germ (the rule of Table 2 with lengths $2, \dots, 10$, completion required for $2, \dots, 12$) contains 38 of its 39 transitions, and shrinking it requires solving exactly the hard subproblems; nogoods learned from the completion problem, as in a logic-based Benders decomposition, would therefore be almost as specific as the germs themselves.

Size of the search. We estimated the size of the natural look-ahead search tree—half-line transitions fixed in order of first use, anti-diagonal by anti-diagonal, a node being discarded when the full formula with the partial table as assumptions is refuted within 3000 conflicts— with Knuth’s random-probe estimator. For four states (MT_4 with lengths $2, \dots, 9$) the mean over 30 probes is 27 nodes. For five states ($MT_5(\{2, \dots, 12\}; 78, 39)$ with links) three probes gave $4.2 \cdot 10^{11}$, $3.1 \cdot 10^{17}$ and $8.3 \cdot 10^{18}$; adding the lengths up to 18 prunes more strongly along some paths but not in general (three probes: $2.2 \cdot 10^5$, $4.2 \cdot 10^{12}$, $1.0 \cdot 10^{15}$). The estimator has a large variance, but the gap of at least ten orders of magnitude is robust and explains why our runs on the unrestricted five-state formulas with the pumping constraints for $N = 12, 14$ and 18 (up to about 1.5 CPU-hours each) did not terminate. Clause learning compresses such trees considerably, and cube-and-conquer [27] parallelizes them almost linearly, but a complete five-state search along these lines is a computation of a much larger scale than the ones reported here.

7. Discussion

What the pumping lemma buys. Lemma 3.3 turns an infinite family of requirements on the reflected triangles into constraints on the half-line

diagram only. It is satisfied by Mazoyer’s solution, as it must be, and it eliminates, at negligible cost, the whole class of rules whose half-line becomes periodic behind the front too early. Theorem 3.4 shows that the constraint has a transparent geometric meaning: in every minimal-time solution the non-periodic part of the half-line reaches the line $i = t/3$. For Mazoyer’s solution every depth-row is 3-periodic from time $2j + 1$ or $2j + 2$, i.e. the periodic zone behind the front is bounded by a signal of speed $1/2$, comfortably within the bound.

Why lucky rules are hard to kill. Small sets of necessary conditions are typically met by rules that are not solutions; the question is how many lengths are needed before all such rules disappear (if no solution exists, some finite set of lengths suffices, since there are finitely many rules). For four states the answer is 9 (or 8 with a long half-line). For five states the rule space is larger by roughly 38 orders of magnitude (5^{94} versus 4^{45} transition tables), and our experiments show rules for all lengths up to at least 13, and partial rules for any single additional length we tried. Under the pumping constraints the surviving rules are of two kinds. Either their half-lines consist of finitely many periodic regions separated by signals, with a boundary of speed $1/3$ (as for δ_{12} and δ_{13}); such half-lines are periodic in a wedge at the left border and are excluded by Theorem 3.6, although only for long lines. Or their half-lines are chaotic (as for δ'_{13}); these pass both barriers, and at present only completion tests on longer lines can exclude them.

The mirrored construction. The cone of Lemma 3.2 ends where the anti-diagonal $t + i = 2n - 2$ meets the line $i = t/3$. By Lemma 3.1, the part of R_n whose cone lies in the periodic zone behind the front is the same for all large n of a given residue class, and the information that makes the right end fire exactly at time $2n - 2$ enters R_n near that crossing and travels to the right end at maximal speed. Heuristically, the right half of every reflected triangle therefore has to behave like a *mirrored* minimal-time synchronization, started by the reflection and running on the background left behind the front instead of on quiescent cells. A five-state solution would have to realize both constructions—the direct one on the quiescent background and the mirrored one on the front’s background—with the same four working states. We regard this tension as the structural reason why five states are hard; turning it into a theorem (for instance a mirrored version of Theorem 3.4, for which the missing ingredient is a border at the far end of the mirrored problem) is the most promising route to a conceptual proof.

The left half. The same time budget applies on the left. The reflected wave reaches cell i at time $2n - 1 - i$ and every cell fires at time $2n - 2$; information created behind the wave cannot overtake it, so the cells to the left of the midpoint can only combine what the half-line has stored around them with what a bounded number of chains immediately behind the wave bring from the right. Theorem 3.6 turns this into a theorem: the half-line cannot be periodic in a wedge at the left border. In the classical constructions the required structure is the hierarchy of slow signals whose crossings with the wave mark the recursive division points; the common half-line of the rules of Theorem 6.1 has none, and Corollary 6.2 excludes it. The bound of Lemma 3.5 is exponential in s and linear in the period, so the lemma is weak for short lines; a sharper version, or a version for half-lines that are only *eventually* periodic along the anti-diagonals, would strengthen the search of Section 6 considerably.

Open problems.

- (1) Does a five-state minimal-time solution exist? By Theorem 6.1, a negative answer along the lines of Theorem 5.1 requires lines of length at least 14.
- (2) Is there a necessary condition, in the spirit of Lemma 3.3, that constrains the reflected triangles themselves and not only the half-line?
- (3) What is the smallest N such that some five-state rule is $\{2, \dots, N\}$ -minimal but no five-state rule is $\{2, \dots, N + 1\}$ -minimal, if such an N exists? We know $N \geq 13$.
- (4) Can the bound $(k - 1)^{2s}L$ of Lemma 3.5 be replaced by one that is polynomial in s , or by one that also covers anti-diagonals that are periodic only up to a bounded number of defects? Either would make the lemma effective for the self-similar half-lines that survive the filters of Section 6.3.

Data availability

All programs, formula generators, rule tables and experiment logs are available in the accompanying repository (directory `fssp5/`). The LRAT certificate of Theorem 5.1 (177 MB) is not stored; it is regenerated deterministically in under a minute by the commands of Appendix B, and its SHA-256 digest is listed there.

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Appendix A. The rules of Theorem 6.1 and Proposition 6.3

Appendix B. Reproducibility

The directory `fssp5/` of the accompanying repository contains all programs, the rule tables and the logs. The main components are:

`src/fsspcheck.c`

independent brute-force simulator: reads a rule table and checks minimal-time synchronization for a range of lengths (full simulation of every line, no use of the agreement lemma);

`src/gencnf.py`

CNF generator for $MT_k(\mathcal{N}; M, N')$ with the options of Section 4 (return chains, pumping, links, symmetry breaking) and the restricted classes used in the experiments;

left	centre	right neighbour				
		*	L	G	A	B
*	L	—	A	A	A	A
L	L	L	L	L	B	A
G	L	A	G	G	L	A
A	L	B	A	B	A	A
B	L	G	B	L	A	A
*	G	—	A	A	A	A
L	G	B	B	G	L	B
G	G	G	G	G	A	B
A	G	A	L	A	A	G
B	G	B	G	L	A	A
*	A	—	G	B	F	L
L	A	A	L	A	A	B
G	A	B	L	A	B	A
A	A	F	B	G	F	A
B	A	A	B	A	G	A
*	B	—	A	A	A	A
L	B	G	B	L	B	B
G	B	A	A	B	B	G
A	B	L	B	A	A	G
B	B	G	A	B	B	L

Table A.4: The five-state rule δ_{12} of Theorem 6.1: entry $\delta(\text{left}, \text{centre}, \text{right})$. The entries $\delta(L, L, L) = \delta(L, L, *) = L$ are the quiescence conditions; “—” marks the unused neighbourhoods $(*, y, *)$. Machine-readable copy: **results/uniformA_2-12.txt**.

left	centre	right neighbour				
		*	L	G	A	B
*	L	–	L	G	L	G
L	L	L	L	L	A	G
G	L	B	G	L	B	A
A	L	A	A	B	A	A
B	L	A	A	L	B	B
*	G	–	A	G	A	G
L	G	B	B	G	B	L
G	G	A	G	G	B	B
A	G	A	L	B	A	A
B	G	F	G	L	G	L
*	A	–	G	B	F	F
L	A	A	L	G	L	G
G	A	A	L	A	A	A
A	A	F	G	A	F	L
B	A	G	A	B	A	L
*	B	–	A	G	G	G
L	B	G	B	L	G	G
G	B	L	B	A	A	A
A	B	F	B	F	A	B
B	B	A	B	A	L	G

Table A.5: The five-state rule δ_{13} of Theorem 6.1: entry $\delta(\text{left}, \text{centre}, \text{right})$. The entries $\delta(L, L, L) = \delta(L, L, *) = L$ are the quiescence conditions; “–” marks the unused neighbourhoods $(*, y, *)$. Machine-readable copy: **results/delta13.txt**.

left	centre	right neighbour				
		*	L	G	A	B
*	L	–	F	F	L	G
L	L	L	L	G	G	B
G	L	G	A	A	L	G
A	L	A	B	A	A	A
B	L	A	B	A	G	L
*	G	–	L	F	F	F
L	G	F	A	F	B	A
G	G	F	B	F	G	F
A	G	G	A	G	G	G
B	G	A	B	L	A	B
*	A	–	F	L	F	F
L	A	A	B	G	A	A
G	A	G	A	A	G	L
A	A	B	L	G	G	L
B	A	B	L	B	L	B
*	B	–	L	F	F	L
L	B	A	B	A	G	L
G	B	F	G	A	A	G
A	B	B	B	G	G	B
B	B	B	L	G	A	L

Table A.6: The five-state rule δ'_{13} of Proposition 6.3: entry $\delta(\text{left}, \text{centre}, \text{right})$. The entries $\delta(L, L, L) = \delta(L, L, *) = L$ are the quiescence conditions; “–” marks the unused neighbourhoods $(*, y, *)$. Machine-readable copy: **results/delta13_chaotic.txt**.

`src/pumping.py`
 computation of the cones of Lemma 3.2 by breadth-first search and direct verification of $\text{PUMP}(n, n')$ on a given rule;

`src/fsspdfs.c`
 exhaustive depth-first enumeration of partial rules (Table 1);

`src/mazoyer_from_coq.py`
 extraction of Mazoyer’s rule from the file `autom.v` of Duprat’s Coq development [16];

`src/validate_mazoyer.py`
 soundness test of the encoding;

`src/check_halfline.py`
 direct check that a half-line diagram has no firing cell below a given anti-diagonal;

`src/fixgerm.py`, `src/knuth.py`, `src/benders.py`
 the experiments of Section 6;

`src/germdfs.c`
 enumeration of the half-line germs of bounded complexity (Proposition 6.4);

`src/germenc.py`, `src/germcert.py`
 completion tests of germs (CaDiCaL under assumptions) and a single LRAT certificate for a list of refuted germs;

`src/germext.c`
 extension of the half-line determined by a germ, and the checks of Lemma 3.5 and Lemma 3.3 on it;

`src/germcompare.py`
 the neighbourhoods used by the half-line of a rule below a given anti-diagonal (Theorem 6.1(c)).

The directory `results/germs` contains the germ lists, the logs of the classification and, in `CERTIFICATES.txt`, the digests of the certificates.

Certificate for Theorem 5.1.

```
python3 gencnf.py 4 9 mt4_9.cnf
cadical --lrat --binary=false mt4_9.cnf mt4_9.lrat # UNSAT
lrat-check mt4_9.cnf mt4_9.lrat # VERIFIED
```

SHA-256 digests (each on two lines):

```
mt4_9.cnf          d18d65480e77505193b8f60362badad5
                   e71f8f08505f29befbcd3b2b18cdaabf
mt4_9.lrat         1d511a75ab25587fa4a3476b95ad3f71
                   33c7375ad96661fd22fa598aec6c8092
uniformA_2-12.txt  b62abbcac0653a661cd1bbe344dee9db
                   ba0cddee37448ea16681d436166c3aad
delta13.txt        99dbe526792323b2f77786bcb3dee4fb
                   ee624e2efac5dc8a52c02cb8fb45b891
mazoyer6.txt       4b0717ea7319957c1fdcc887fe09c107
                   4cf338e8b849cdd32b4307d0083f417e
```

Checking Theorem 6.1.

```
R=results/uniformA_2-12.txt
fsspcheck $R 2 12      # OK for all 2 <= n <= 12
fsspcheck $R 13 13     # FAIL: a cell fires at time 23
python3 pumping.py $R 40 # 0 violations
python3 check_halfline.py $R 78 # no firing cell, t+i <= 78
S=results/delta13.txt
fsspcheck $S 2 13      # OK for all 2 <= n <= 13
fsspcheck $S 14 14     # FAIL: a cell fires at time 21
python3 pumping.py $S 40 # 0 violations
python3 check_halfline.py $S 78 # no firing cell, t+i <= 78
python3 germcompare.py $R $S 78 # identical, 22 besides (L,L,L)
```

The uniform configurations before firing in (a) and (b) are visible with the option `-v` of the checker, which prints the diagrams.

Checking Corollary 6.2 and Propositions 6.3 and 6.4.

```
gcc -O2 -o germext germext.c && gcc -O2 -o germdfs germdfs.c
# Corollary 6.2: germ of delta12/delta13 (22 neighbourhoods)
./germext 5 1300 3 6 < ./results/germs/germ_delta12.txt
# CLOSED; REFUTED n=518 s=2 L=1 Y=259
# Proposition 6.3
./fsspcheck ./results/delta13_chaotic.txt 2 13 # OK
./germext 5 10000 3 24 800 < ./results/germs/germ_delta13c.txt
# CLOSED, no refutation
# Proposition 6.4 (complexity <= 14)
./germdfs 5 78 14 40 > germs_K14.txt # 244 germs
python3 germcinc.py germs_K14.txt 10 78 # 230 UNSAT, 14 SAT
python3 germcert.py <the 230 refuted germs> 10 k14_n10 # UNSAT, VERIFIED
./germext 5 3300 3 12 < germs_K14_surv10.txt # 14 x REFUTED (s=1)
```

The files `results/germs/germ_delta12.txt` and `germ_delta13c.txt` contain the two germs in the one-line format of `germdfs.c`, as printed by `germcompare.py -germline`; the complete pipeline for complexity at most 16 is `germpipe.sh`.

Validation against Mazoyer's solution.

```
python3 mazoyer_from_coq.py autom.v > mazoyer6.txt
fsspcheck mazoyer6.txt 3 3000    # OK for all  $3 \leq n \leq 3000$ 
python3 validate_mazoyer.py 14   # SAT under exactly one renaming
```